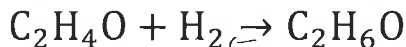


A rigid vessel is initially filled with one mole of a gas mixture of 50 mol% hydrogen (H_2) and 50 mol% acetaldehyde (C_2H_4O) at a total pressure of 760 mmHg absolute and $25^\circ C$. Ethanol (C_2H_6O) forms according to the chemical reaction:



(a) After the system comes to equilibrium at constant temperature of $25^\circ C$, the pressure has dropped to 700 mmHg absolute. Calculate the total moles and the moles of each species present. Assume ideal gas behavior.

(b) If the equilibrium expression is given by $K = \frac{y_{C_2H_4O} y_{H_2}}{y_{C_2H_6O}} \left(\frac{P_0}{P(atm)} \right)$ where $P_0 = 1$ atm and $P(atm)$ is the pressure in atmospheres absolute, what is the value of the equilibrium constant K ?

a) $PV = nRT$

$$760 \text{ mmHg} \approx 1 \text{ atm}$$

$$(1 \text{ atm})(22.4 \text{ L}) = (1)(.0831)(298^\circ K)$$

$$n = 24.7638$$

Use textbook notation $m(n_{C_2H_4O, out})$

$$C_2H_4O: \quad C_2H_4O_{out} = C_2H_4O_{in} - \xi_1 \rightarrow C_2H_4O = .5 - \xi_1$$

$$H_2: \quad H_{2, out} = H_{2, in} - \xi_1 \quad H_{2, out} = .5 - \xi_1$$

$$C_2H_6O: \quad C_2H_6O_{out} = C_2H_6O_{in} + \xi_1 \quad C_2H_6O_{out} = + \xi_1$$

$$n_{tot, out} = 1 - \xi_1$$

After equilibrium

$$(.921 \text{ atm})(24.76) = n (.0831)(298^\circ K)$$

$$n = .9208 \text{ mol} \rightarrow .920 = C_2H_6O_{out} + H_{2, out} + C_2H_4O_{out}$$

$$.9208 = \xi_1 + 1 - 2\xi_1$$

$$.07914 = \xi_1$$

so $C_2H_4O_{out} = .42085 \text{ mol}$

Total moles out = .9208 mol

$$H_{2, out} = .42085 \text{ mol}$$

$$C_2H_6O_{out} = .07914 \text{ mol}$$

b)

$$K = \frac{P_0}{P_{\text{atm}}} \left(\frac{Y_{\text{C}_2\text{H}_4\text{O}}}{Y_{\text{C}_2}} \frac{Y_{\text{H}_2}}{0} \right)$$

$$P_0 = 1 \text{ atm}$$

$$Y_{\text{C}_2\text{H}_4\text{O}} = .420 = Y_{\text{H}_2}$$

$$Y_{\text{C}_2\text{H}_6\text{O}} = .07991$$

$$= \frac{(.420)(.420)}{(.07991)(0.921 \text{ atm})} \cdot 1 \text{ atm}$$

$$K = 2.406$$

put units on values!